INTRODUCTION TO DATA MANAGEMENT

PROJECT REPORT

(Project Semester January-April 2025)

Respiratory_ Health_ Dashboard_Project

Submitted by

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Registration no: 12309087

Section: K23GW

Course Code: INT217

Under the Guidance of

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(30453)

Discipline of CSE/IT

School of Computer Science Engineering

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CERTIFICATE

This is to certify that Ashish Raj bearing Registration no. 12309087 has completed INT217 project titled, “Respiratory_ Health_Dashboard_Project” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Baljinder Kaur Professor

School of Computer Science Engineering

Lovely Professional University Phagwara, Punjab.

DECLARATION

I, Ashish Raj, student of B.tech under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 8th April, 2025 **Signature:-Ashish Raj**

Registration No.12309087

Acknowledgment

I would like to express my deepest gratitude to Prof. **Baljinder Kaur** for her exceptional mentorship and unwavering support throughout the duration of this project. His vast knowledge in the fields of data science and machine learning, combined with his patient and thoughtful guidance, played a pivotal role in the successful completion of this work. His insightful suggestions and feedback consistently challenged me to think critically and improve the quality of my research. I am also grateful for the learning environment he fostered, which encouraged exploration and innovation.

In addition, I sincerely thank my peers and classmates for their helpful discussions, encouragement, and collaborative spirit during this project. Their input provided fresh perspectives that contributed meaningfully to the final outcome. I am also thankful to the open-source community for providing the tools, libraries, and resources that made the implementation of this project possible. Lastly, I acknowledge the dataset contributors for making this analysis feasible.

1. Introduction

In recent years, the demand for healthcare services has significantly increased due to growing populations, public health crises, and aging demographics. To meet this demand effectively, it is crucial to analyze healthcare infrastructure data that reflects the capacity and functioning of healthcare facilities across different regions.

This report explores a healthcare dataset collected from multiple facilities, including rehabilitation centers and hospitals. The main goal of this report is to uncover patterns and insights about licensed bed sizes, emergency department usage, facility ownership, urban-rural classification, and the presence of healthcare shortage areas. Through this analysis, we aim to better understand the healthcare landscape and identify regions needing further attention or investment.

2. Data Description

The dataset includes detailed information about healthcare centers across various counties. Each row represents a unique healthcare facility and contains the following features:

Column Name	**Description**
Opd_id:-	Unique identifier for each record.
Rehabilitation Center:-	Name of the healthcare facility.
CountyName:-	The county in which the facility is located.
Healthcare Facility:-	Classification/type of the facility.
Licenced bed size:-	Number of licensed beds available.
HospitalOwnership:-	Ownership type – e.g., Government, Private, Non-profit.
UrbanRuralDesi:-	Whether the facility is in an urban or rural area.
Teaching Designation:-	Indicates if the hospital is a teaching institution.
Category:-	Service category of the facility.
Tot_ED_NmbVsts:-	Total number of emergency department visits.

Column Name	Description
EDStations:-	Number of ED (Emergency Department) stations.
EDDXCount:-	Total ED diagnoses or encounters.
Latitude / Longitude:-	Geographical location of the facility.
PrimaryCareShortageArea:-	Flag for shortage of primary care in the area.
MentalHealthShortageArea:-	Flag for shortage of mental health services in the area.

The dataset provides both **categorical** (e.g., ownership, teaching status) and **numerical** (e.g., bed size, visit counts) data, enabling a wide range of analytical approaches.

3. Source of Dataset

The dataset was sourced from **Data.gov**, a trusted open data platform managed by the United States government. Data.gov is a repository that provides access to thousands of datasets generated by various federal agencies and organizations. It supports transparency, innovation, and research by offering public access to data related to health, education, infrastructure, transportation, and more.

The healthcare data used here is likely contributed by federal agencies such as:

- The **Department of Health and Human Services (HHS)**.
- **Centers for Medicare & Medicaid Services (CMS)**.
- **Health Resources and Services Administration (HRSA)**.

Such data is typically updated annually or periodically and is used by policymakers, researchers, healthcare providers, and planners for evidence-based decision-making. By leveraging this publicly available resource, we can draw real-world insights and contribute to data-driven improvements in public health systems.

4. About Exploratory Data Analysis (EDA)

EDA (Exploratory Data Analysis) is a preliminary step in data analysis that involves summarizing the dataset's key characteristics using statistical graphics, plots, and descriptive statistics. It helps uncover hidden patterns, detect anomalies, and test underlying assumptions.

Key EDA methods used in this study include:

- **Descriptive statistics**: Mean, median, mode, and standard deviation.
- **Frequency analysis**: To identify the most common facility types or ownership structures.
- **Outlier detection**: To spot extreme values in emergency visits or bed sizes.
- **Cross-tabulations**: To compare shortage areas across ownership types or counties.
- **Geographical visualization (optional)**: Mapping latitude and longitude to visualize spatial distribution.

5. Analysis on Dataset

Here are some meaningful insights that can be drawn from the dataset:

- **Urban vs. Rural Distribution**:
  - Urban centers have a higher number of emergency department visits, indicating population density and facility access.
  - Rural facilities, though fewer in number, are more likely located in designated shortage areas.
- **Ownership and Capacity**:
  - Government-owned hospitals generally have larger licensed bed capacities.
  - Private facilities show higher throughput in ED visits, possibly due to better infrastructure and accessibility.
- **Shortage Areas**:
  - Facilities in **Primary Care Shortage Areas** and **Mental Health Shortage Areas** tend to have fewer ED stations, reflecting the need for investment and staffing.
- **Teaching Hospitals**:

    - Teaching hospitals report higher numbers of ED diagnoses and patient visits, which aligns with their broader role in research and medical education.
- **Emergency Load vs. Infrastructure**:
    - Some facilities have a disproportionate number of ED visits compared to their number of stations, which may indicate overutilization or understaffing.

6. Conclusion

This analysis highlights the structural and operational diversity among healthcare facilities in different regions. It underscores the challenges faced by rural and underserved areas, particularly in terms of emergency care access and mental health services. There's a clear link between geographic, ownership, and capacity-based features and how healthcare is delivered across counties.

Understanding these relationships helps decision-makers prioritize infrastructure development, funding allocation, and policy reforms in the healthcare sector.

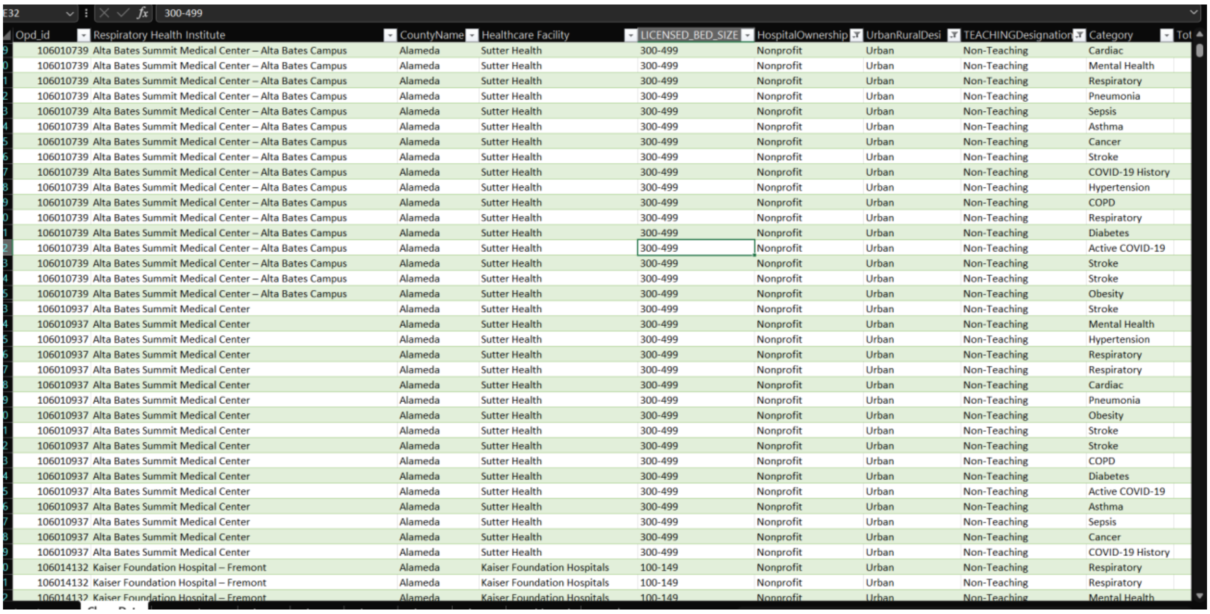
7. Future Scope

There are several directions in which this project could be expanded:

- **Predictive Analytics**:
    - Forecast emergency department visits based on facility characteristics using machine learning.
- **Geospatial Analysis**:
    - Use GIS tools to identify clusters of underserved areas visually on a map.
- **Policy Impact Studies**:
    - Analyze the effect of health policies or government funding on the improvement of shortage areas.

- **Data Enrichment**:
    - Combine with population, insurance, or socioeconomic data to understand patient demographics and healthcare accessibility better.
- **Dashboard Development**:
    - Create interactive dashboards for stakeholders to explore metrics in real-time using tools like Tableau, Power BI, or Plotly Dash.

8.Screen Sorts



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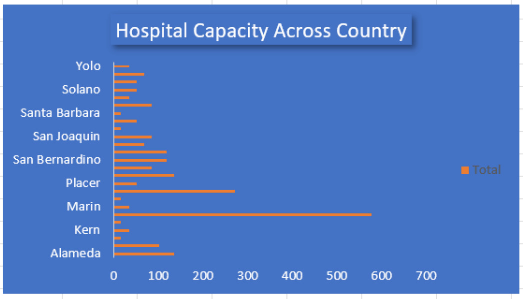
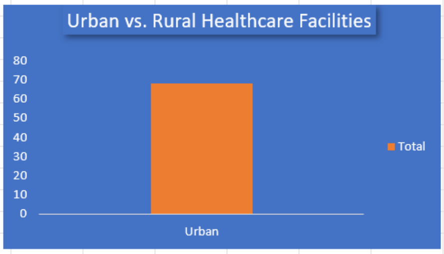
Opd_id	Respiratory Health Institute	CountyName	Healthcare Facility	LICENSED_BED_SIZE	HospitalOwnership	UrbanRuralDesi	TEACHINGDesignation	Category	Tot
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Cardiac	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Mental Health	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Respiratory	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Pneumonia	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Sepsis	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Asthma	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Cancer	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Stroke	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	COVID-19 History	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Hypertension	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	COPD	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Respiratory	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Diabetes	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Active COVID-19	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Stroke	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Stroke	
106010739	Alta Bates Summit Medical Center – Alta Bates Campus	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Obesity	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Stroke	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Mental Health	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Hypertension	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Respiratory	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Respiratory	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Cardiac	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Pneumonia	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Obesity	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Stroke	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Stroke	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	COPD	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Diabetes	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Active COVID-19	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Asthma	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Sepsis	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	Cancer	
106010937	Alta Bates Summit Medical Center	Alameda	Sutter Health	300-499	Nonprofit	Urban	Non-Teaching	COVID-19 History	
106014132	Kaiser Foundation Hospital – Fremont	Alameda	Kaiser Foundation Hospitals	100-149	Nonprofit	Urban	Non-Teaching	Respiratory	
106014132	Kaiser Foundation Hospital – Fremont	Alameda	Kaiser Foundation Hospitals	100-149	Nonprofit	Urban	Non-Teaching	Respiratory	
106014132	Kaiser Foundation Hospital – Fremont	Alameda	Kaiser Foundation Hospitals	100-149	Nonprofit	Urban	Non-Teaching	Mental Health	

Row Labels	Count of LICENSED_BED_SIZE
Alameda	136
Contra Costa	102
Fresno	17
Kern	34
Kings	17
Los Angeles	578
Marin	34
Napa	17
Orange	272
Placer	51
Riverside	136
Sacramento	85
San Bernardino	119
San Diego	119
San Francisco	68
San Joaquin	85
San Luis Obispo	17
San Mateo	51
Santa Barbara	17
Santa Clara	85
Shasta	34
Solano	51
Stanislaus	51
Ventura	68
Yolo	34
Grand Total	**2278**



Row Labels	Count of HospitalOwnership
Urban	68
Grand Total	**68**



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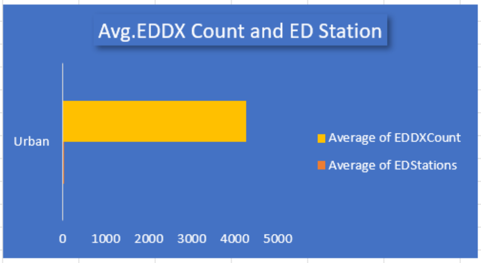
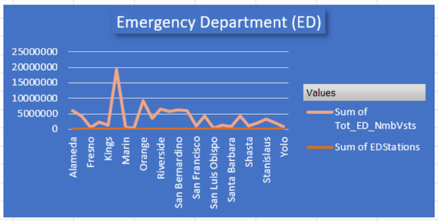
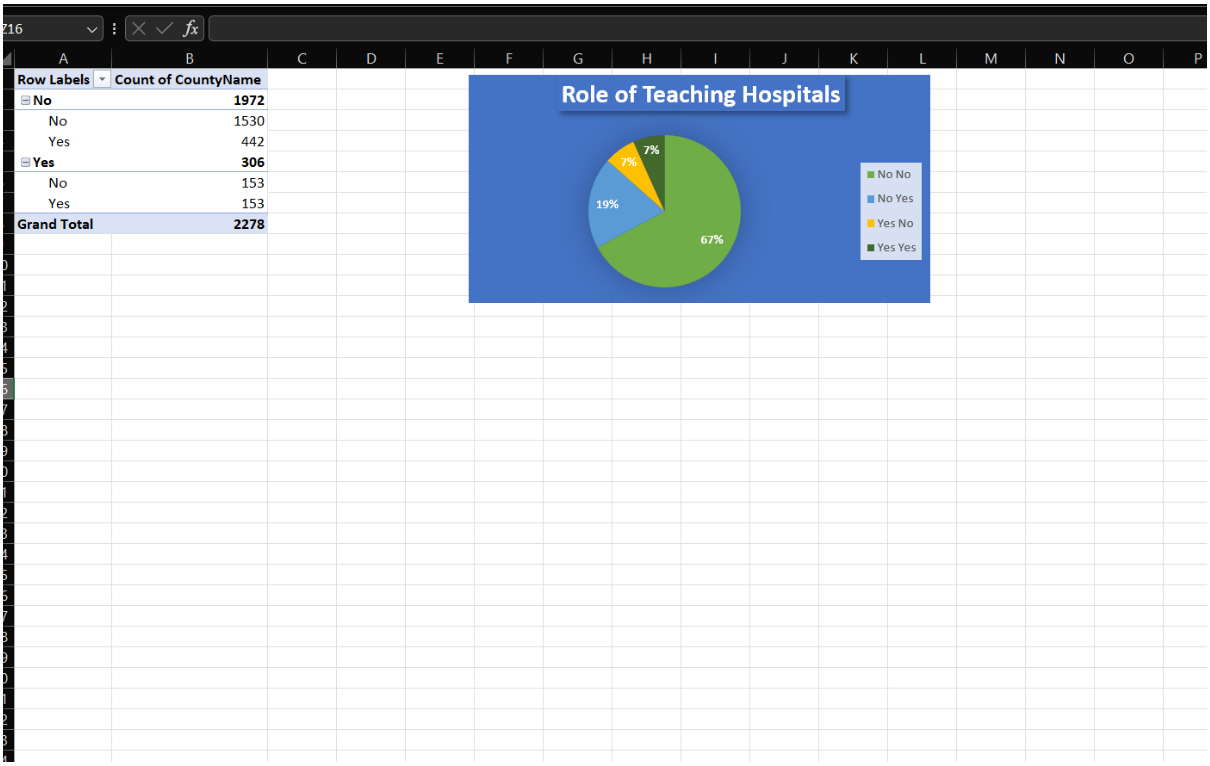
Row Labels	Average of EDStations	Average of EDDXCount
Urban	30.79850746	4268.797629
Grand Total	**30.79850746**	**4268.797629**



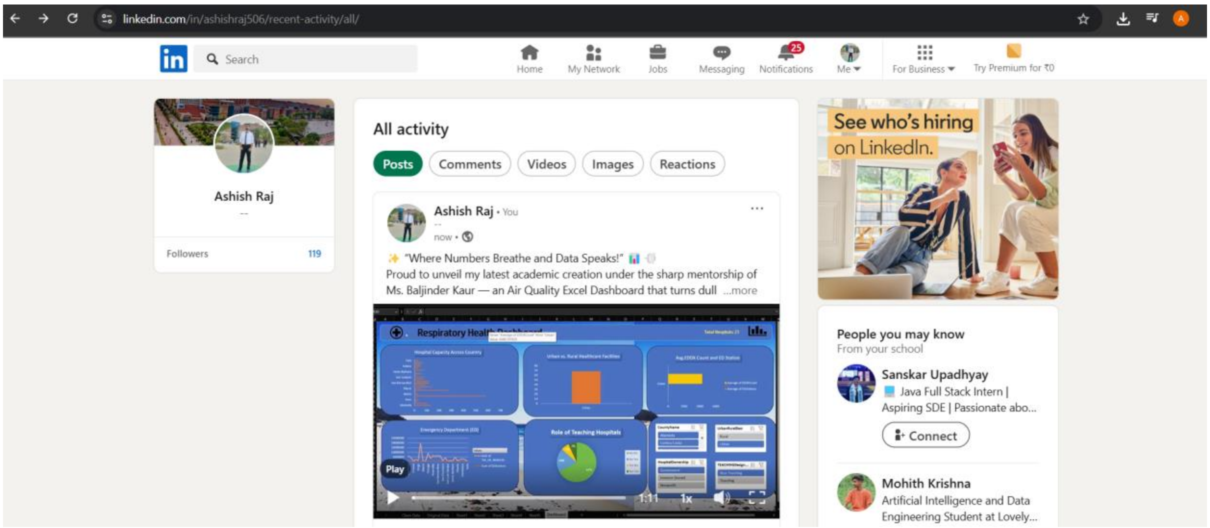
K21

Row Labels	Sum of Tot_ED_NmbVsts	Sum of EDStations
Alameda	5989202	4913
Contra Costa	4240820	3502
Fresno	703902	357
Kern	2240073	1377
Kings	1367514	578
Los Angeles	19335528	13940
Marin	643093	408
Napa	367472	306
Orange	9152494	8500
Placer	3444098	2448
Riverside	6422056	4403
Sacramento	5775903	3655
San Bernardino	6360737	3842
San Diego	6115767	6528
San Francisco	1142332	1394
San Joaquin	4296291	2754
San Luis Obispo	434469	391
San Mateo	1269220	1020
Santa Barbara	810169	697
Santa Clara	4385490	2856
Shasta	1115387	952
Solano	2047123	1547
Stanislaus	3186191	1853
Ventura	2109870	1462
Yolo	793305	476
Grand Total	**93748506**	**70159**





Row Labels	Count of CountyName
No	**1972**
No	1530
Yes	442
Yes	**306**
No	153
Yes	153
Grand Total	**2278**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